

Contents

- [Problem 8](#)
- [Problem 11](#)
- [Problem 12](#)
- [Problem 13](#)
- [Problem 14](#)
- [Problem 15](#)
- [Problem 17](#)

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% ECE 341 Lab 1 - Introduction to MATLAB                                %
% Author: Gianni Avilan                                              %
% 02.04.2026                                                         %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

clear          % clear existing variables from the workspace
close all      % close all figures
clc            % clear the command window of text
```

Problem 8

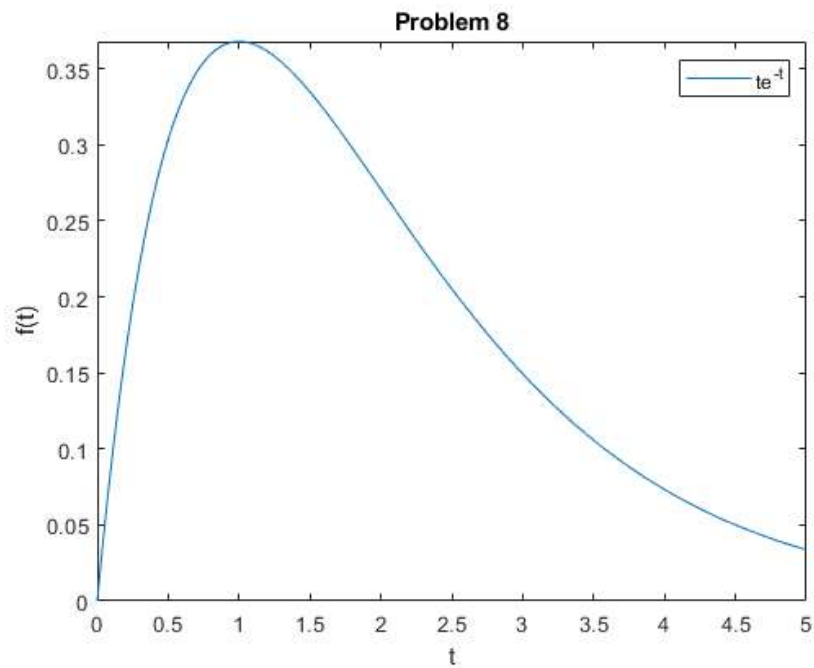
```
clear
close all
clc

fprintf('Problem 8:\n\n');
% generate time vector
t = [0:.001:5];
% generate f(t) vector
f = t.*exp(-t);
% plot function, set parameters
figure();
plot(t,f);
title('Problem 8');
xlim([0,5]); ylim([min(f), max(f)]);
xlabel('t'); ylabel('f(t)');
legend('te^{-t}');

fprintf('See displayed figure.\n\n');
```

Problem 8:

See displayed figure.



Problem 11

```
clear
close all
clc

fprintf('Problem 11:\n\n');
% generate A matrix
A = [1:3; 4:6; 7:9];
% create column vector from values of A, display output
fprintf('The column vector of A, row by row, is\n');
a = [A(1,:) A(2,:) A(3,:)]'
```

Problem 11:

The column vector of A, row by row, is

```
a =

     1
     2
     3
     4
     5
     6
     7
     8
     9
```

Problem 12

```
clear
close all
clc

fprintf('Problem 12: \n\n');
% generate two random complex numbers%
function[z_3, z_4] = complex_mult_div(z_1, z_2)
    % multiply/divide
    z_3 = z_1*z_2; z_4 = z_1/z_2;
    % print results
```

```

    fprintf('The product of %i + j%i and %i + j%i is %i + j%i\n', real(z_1), imag(z_1), real(z_2), imag(z_2), real(z_3), imag(z_3));
    fprintf('The division of %i + j%i and %i + j%i is %.i + j%.i\n\n', real(z_1), imag(z_1), real(z_2), imag(z_2), real(z_4), imag(z_4));
end

%test the function
fprintf('Testing the function,\n\n');
fprintf('Providing two complex numbers, z1 = -3+2j and z2 = 2-4j,\n\n');
complex_mult_div(-3+j*2, 2-j*4);

```

Problem 12:

Testing the function,

Providing two complex numbers, z1 = -3+2j and z2 = 2-4j,

The product of -3 + j2 and 2 + j-4 is 2 + j16

The division of -3 + j2 and 2 + j-4 is -7e-01 + j-4e-01

Problem 13

```

clear
close all
clc

fprintf('Problem 13: \n\n');
function[max_value] = absolute_max(x1, x2)
    %put numbers in array, abs() them, return max from the array
    value_array = [x1, x2];
    abs_array = abs(value_array);
    max_value = max(abs_array);
    fprintf('The highest absolute value between %f and %f is %f\n', x1, x2, max_value);
end

%test the function
fprintf('Testing the function,\n\n');
fprintf('Providing two numbers, x1 = -3 and x2 = 2,\n\n');
absolute_max(-3,2);

fprintf('Providing three numbers, x1 = -3, x2 = 2, x3 = 5, returns an error,\n\n');

```

Problem 13:

Testing the function,

Providing two numbers, x1 = -3 and x2 = 2,

The highest absolute value between -3.000000 and 2.000000 is 3.000000

Providing three numbers, x1 = -3, x2 = 2, x3 = 5, returns an error,

Problem 14

```

clear
close all
clc

fprintf('Problem 14: \n\n');
function[descending_vector, zero_elements_count, non_zero_vector] = vector_manipulation(input_vector)
    fprintf('The given vector, sorted into descending order, is\n');
    % use built-in sort function
    descending_vector = sort(input_vector,'descend')
    % init non-zero vector
    non_zero_vector = zeros(1,length(input_vector));
    % for loop through vector to count 0 elements
    zero_elements_count = 0;

```

```

for i=1:length(input_vector)
    if input_vector(i) == 0
        zero_elements_count=zero_elements_count+1;
        % we can also fill the non-zero vector at the same time
    else
        non_zero_vector(i) = input_vector(i);
    end
end
fprintf('The number of zero elements in this vector is %f\n\n', zero_elements_count);
fprintf('The vector of non-zero elements from the input vector is\n');
non_zero_vector
end

% test the function
fprintf('Testing the function,\n\n');
fprintf('Providing a vector a = [0 0 2 4 5 -4 5 2],\n')
a = [0 0 2 4 5 -4 5 2];
vector_manipulation(a);

```

Problem 14:

Testing the function,

Providing a vector a = [0 0 2 4 5 -4 5 2],
 The given vector, sorted into descending order, is

descending_vector =

```

5    5    4    2    2    0    0   -4

```

The number of zero elements in this vector is 2.000000

The vector of non-zero elements from the input vector is

non_zero_vector =

```

0    0    2    4    5   -4    5    2

```

Problem 15

```

clear
close all
clc

fprintf('Problem 15: \n\n')
% generate vector for primes
nums = 3:101;
% init output vector, add 3 (won't show up in the for loop)
output_vector = [3];
% check primes using remainders
for i= 1:length(nums)
    remainder = rem(nums(i),6);
    if (remainder == 1 || remainder == 5)
        % add primes to output
        output_vector = [output_vector, nums(i)];
    end
end
fprintf('All primes between 3 and 101 are contained in the following vector:\n')
output_vector

```

Problem 15:

All primes between 3 and 101 are contained in the following vector:

output_vector =

Columns 1 through 13

3 5 7 11 13 17 19 23 25 29 31 35 37

Columns 14 through 26

41 43 47 49 53 55 59 61 65 67 71 73 77

Columns 27 through 34

79 83 85 89 91 95 97 101

Problem 17

```
clear
close all
clc

fprintf('Problem 16: \n\n');
% set up the symbolic expressions
syms x y
eqn1 = y == 1 - x^2; eqn2 = y == 1 + x;
% solve
sol = solve([eqn1, eqn2], [x,y]);
% display solutions
fprintf('The solution to the system of equations y = 1-x^{2}; y=1+x is:\n');
fprintf('x = %i; y=%i\n', sol.x(1), sol.y(1));
```

Problem 16:

The solution to the system of equations $y = 1-x^2$; $y=1+x$ is:
 $x = -1$; $y=0$